

Testing the Brenner's theory of transition from feudalism to capitalism

The results of the Brenner Debate produced an enhanced theory of transition from feudalism to capitalism based in rural 17th Century England. A key tenant of this school of thought is the role of "improvement" as both a motor in the divorce between peasants and their means of subsistence, and a consequence of newly created market imperatives both landlords and tenants were now subject too. To test the role of improvement ideology in the transition an agent based model is constructed. This incorporates tenants, landlords, ecological deterioration and rental markets. The critical role of rental market pressure is modelled through a transition from customary rent to leasehold rents. Specific attention is paid to whether increasing tenant mobility and spread of improvement ideology could explain the emergent behaviour of large landlord wealth disparity and consolidated large scale tenancies which characterised agrarian capitalism.

1 Sub-research questions

- Can we test the internal consistency of the model. Given its assumption does it produce the predicted results.
- Can I show the specificity of english agricultural capitalism. ie that my model only transitions under very specific conditions which correspond to the history specificity of england
- How to model the spread of the market imperative as a network where there is an initial point of competition
- How did land ownership and concentration evolve over this transition?
- According to what parameters do the dynamics change rapidly? Its a multi level selection process of both tenants and landlords
- What are the dynamics of tenant productivity, is there a group selection for those landlords with the highest average land productivity(most capital reinvestment)?
- What does the level of productivity scarcity trigger in the model?
 - Does a transition occur within a given time frame if the initial scarcity is low enough?
- Is capital reinvestment a necessity to survive, under what conditions does improvement ideology not spread?
 - What role does variable tenant mobility have?
- How does the size and wealth of landlords evolve given the rental market and heterogeneous tenant productivity?
- Model would focus on a rental market that has bounded rationality (most likely through limited knowledge) which spreads overtime to various agents. How does the bounded rationality aspect affect dynamics?

2 Questions to answer from literature review

1. What causes customary rents to switch to leaseholds? its decreasing income of the landlord, so i just say if the landlord can pay for his stuff he starts switching customary rents to leaseholds
2. How do i model inflation? just make it exogenous?
3. How do i make the model not deterministic? how do i make sure its not a choice
4. How does making the propensity to re-invest into land like a variable that is linked to each individual? is this accurate, at the end i am spreading an ideology, no?
5. how do i use a replicator model in this? is it relevant?
6. How do i code the market aspect in Both for the mobility of tenants and the rent market
7. Who was improving the land, like who was investing their wealth into it? Was it the landlords or the renters
8. How do yeomen, leaseholders and those with customary rights fit in together?

9. How did the commons affect all this system? Especially the survival of rentiers and yeomanry
10. how do i nail down the specificity of agrucultural capitalism
11. What was the driving force for the collapse of feudalism
 - why cant it sustain itself?
 - why does it evolve from a more direct method of expropriation through threat of force to a more rent based economy?
 - dans opinion : ecological collapse - malthusian but i cant remeber where i read this/stole it from
12. How were the commons managed, by how much did the amount of land assigned to the commons change over the course of the transition and how much of an additional drive was this towards the proletarianisation of the peasants. Seems wise to leave this out at first, and add later, trying to replicate the 2nd question and potentially answer the 3rd.

3 Stylised facts

- My understanding of the model i want to make is it starts off with lots of tenants and lots of land lords, however there are long tenures, the land quality decreases over time and landlords force tenants off their customary tenures onto leasehold short tenures when their incomes fall below their consumption levels partly due to inflation and decreasing agriculture returns
- In turn the tenants who have surplus reinvest to improve their lands so that they can keep up with the rent
- It makes sense to have some 'ease of eviction' attribute for landlords, determining how readily they can evict tenants who fail to always pay their rents. This quantity which will be much higher for estates with market rents than those with customary rents (on top of the fact that tenants will have a harder time paying market rents). Given that improvement was used as a legal basis for enclosures and evictions (NEED REFERENCES), it would make sense to have a landlord's adoption of improvement ideology - and its prevalence in society - heavily influence the ease of eviction.

4 Dan's model

4.1 What are the essential elements

- ecological collapse - feudalism by itself cant survive. (malthusian cycles)
- falling profits of landlords - feudalism by itself cant survive
- land based view - there has to be a relation between where agents are and their interactions
- domestic demand for goods
- enclosure - people losing access to their means of self reproduction
- agents act not to impose a transition but to maintain their status and wealth, their class
- primitive accumulation as a change in social relations not wealth
- trifecta of landlord, capitalist tenant and wage labourer
- market dependency of land access - evolution towards this
- market dependency of goods - evolution towards this

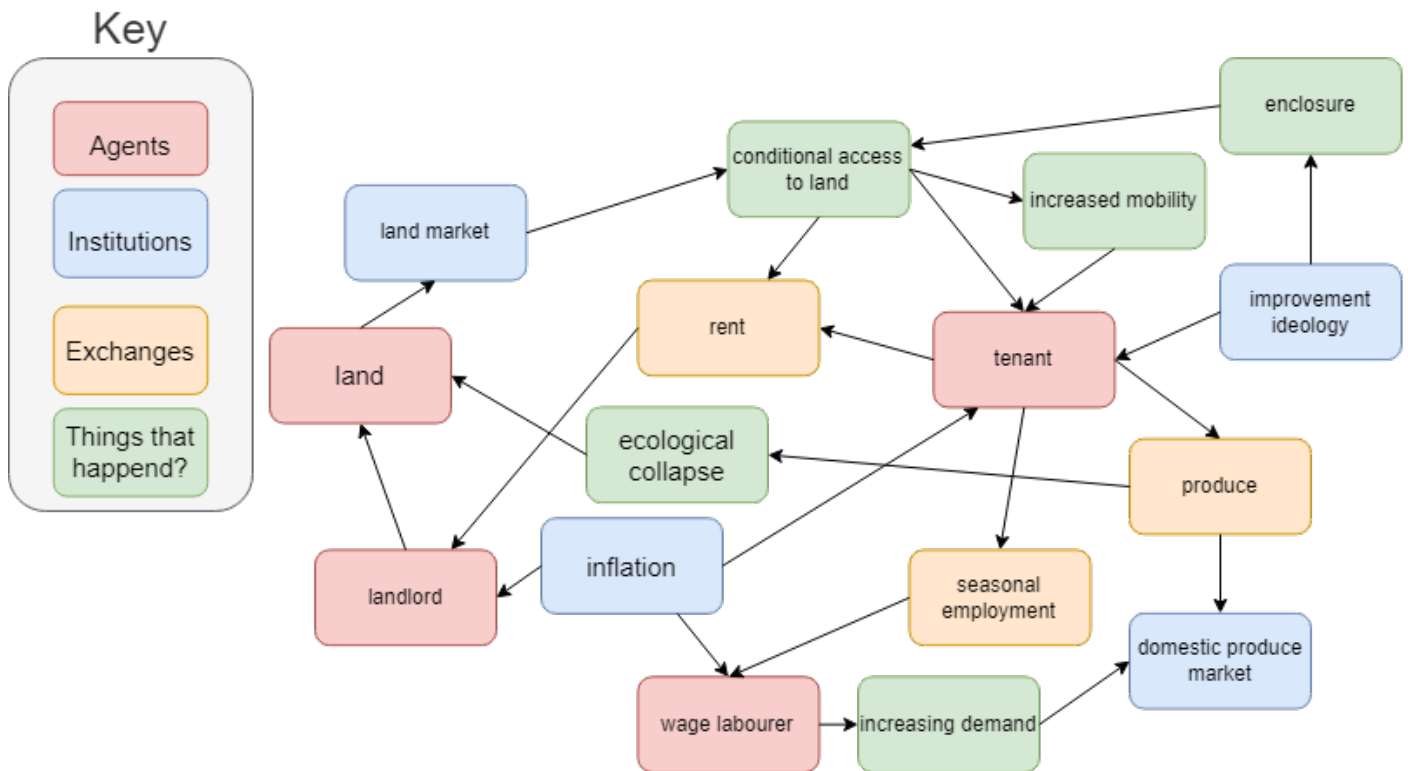


Figure 1: Model structure

4.2 Diagram connections

- Landlord to land : Landlords own land that is leased out to tenants. These landlords set rents for the land
- Land to land market: Land market requires land available to be traded
- Land market to conditional access to land: Access to land requires a land market to be able to rent available land.
- Conditional access to land to rent: the more scarce the access to land the higher the rent[CITE]
- Rent to landlord: Landlords gain income from rent of land
- ecological collapse to land: The repeated use of land with medieval methods leads to decreasing agricultural yields and cyclical collapse - malthus[CITE]
- inflation to landlord: Inflation over time lowers[CITE] the fixed income of landlords who historically have a majority of customary rents[CITE]
- Inflation to wage labourer: ???
- Seasonal employment to wage labourer: Wage labourer existed prior to the dominance of capitalism in the form of short term contracts to deal with spikes in labour activity such as harvest[CITE]. The wage labourers during this period would be working on land that was not theirs but might still have access to land of their own. This was not their sole source of income.[CITE]
- Inflation to tenant: Reduces purchasing power over time and access to essential goods. Less sure about this.
- Wage labourer to increasing demand: Lack of access to land means no method to feed oneself hence have to fully commit to wage labour to earn enough to buy essential goods.
- Increasing demand to domestic produce market: Increased demand from wage labourers for essential goods produces market whereby this demand now affects production. How is this different to dutch importation of wheat.

5 (Ben's) understanding of Wood's model (from OoC chaps 5, 6)

5.1 Wood's take on what marks the capitalist mode of production apart

The type of appropriation of surplus value (economic means only); market dependence, which brings with it the imperatives of competition, accumulation, profit-maximisation "and hence a constant systemic need to develop the productive forces". Therefore the capitalist part of the economy can and must expand.

So perhaps the most important attribute to determine whether our model validates the theory is whether production (per unit labour [and per unit land]?) ends up consistently increasing. Perhaps the fraction of production for exchange vs production for direct consumption, as tenant farmers are gradually put into increasingly precarious positions, may also be used as a proxy for the expansion of the proto-capitalist sector of the economy - not sure how within the scope of the model this is though.

5.2 Wood's Initial Conditions

- The English state was more centralised - "cohesive" - than many others, partly as a result of the Norman conquest of the 11th century, leading to decreased autonomy for Feudal lords, and thus decreased avenues for extracting surplus through extra-economic means.
 - Related - the aristocracy was relatively demilitarised, and so reliant on the state to protect the institution of private property so that they could extract surplus value, increasingly by economic means.
- Already, then, agrarian landlords had a material incentive to compel their tenants to be more productive.
- "Impressive network of roads and water transport", providing the necessary infrastructure for the concentration of the population in London and an emerging national market - Wood seems to imply that this was already happening by the 16th century. - Perhaps this one is less important / relevant.
- Land ownership was unusually concentrated in fewer hands, relative to other countries; something of a "trade-off" between landlords' lack of military and direct local political powers and their control/ownership of the land.
 - Therefore, an unusually high proportion of the land was worked by tenants, even before enclosure really took off.
- A growing number of rents were market rents (pre-existing erosion of systems of customary rents) - a market in leases was developing.

5.3 Consequences of ICs and resulting evolution

- Farmers - including prosperous yeomen - were market-dependent for their access to the land.
- In order to pay these rents, tenant farmers had to produce for exchange on the market, which developed further, becoming more integrated over a wider area.
- These farmers, now subject to increasing competition, were now under an imperative to increase their productivity; those less able to compete ultimately lost access to the land and became wage-labourers.
- Hence, we have the increasing dominance of the "Agrarian triad" - landlord, capitalist tenant and wage labourers.
- Even landowners working their own land were subject to the imperatives associated with competition, as those among them who were able and willing to successfully produce for exchange would see their holdings grow; those who were not would be more likely to lose their land in the long run (this is my interpretation of p.103, top paragraph).
- This last aspect of the emerging dynamics was strengthened by the increasing enclosures and evictions that dispossessed the peasantry of their land and forced them into market-dependency for their access to the means of production.
 - Improvement ideology, for Wood, developed largely in the 17th century as a (legal and moral) justification for enclosure and the compulsion for 'producers' to increase their productivity.
- Increased food production led to a large increase in population, increasingly centred in London (London's population went from 60,000 in 1530 to 575,000 in 1700).

- This dispossessed population was dependent on the market not only in terms of its access to the means of production, but to the means of its own self-reproduction - food, but domestic products more generally (Woods names iron cooking pots as an example).
- New forms of banking were developed in London, focused on the “metropolitan market”, whereby bankers would give credit to producers of said domestic products.
- Due to the increased productivity of agricultural labour, an increasing portion of the population could (indeed, had to) take up non-agricultural occupations. This large workforce, coupled with the increasingly nationally integrated consumer market both allowed and, through the imperative to increase productivity, necessitated the development of British industrial capitalism.

Finally, Wood notes that the decline of the English peasantry “may have taken longer than has sometimes been suggested, extending well into the nineteenth century”, but she holds that the increase in market dependence of tenant farmers from the 16th century onward played a great role in cementing this trend.

6 Possible Agents and Attributes

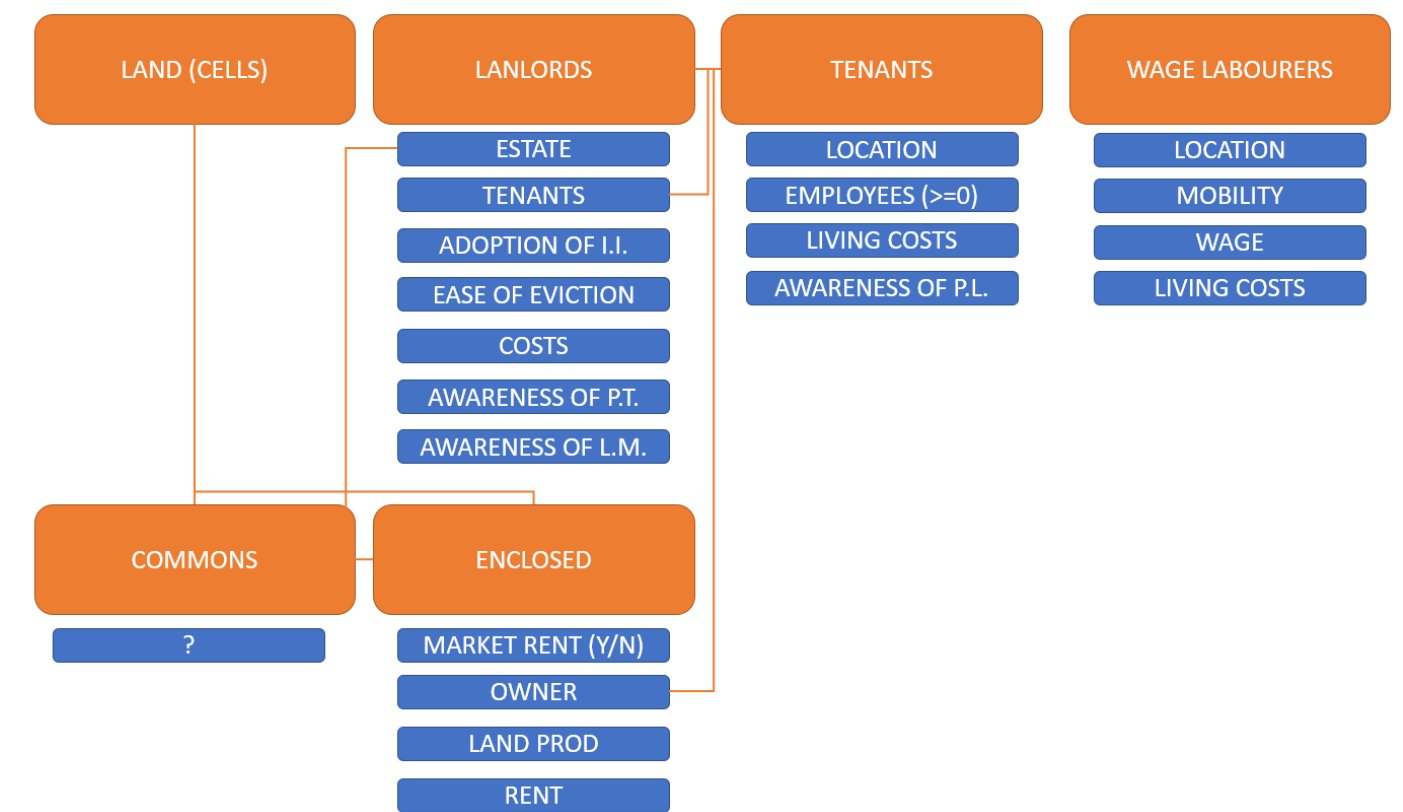


Figure 2: Agents and attributes. Agents are coloured orange, and their attributes blue. P.T. is an abbreviation for potential tenants, I.I. improvement ideology, L.M. land market and P.L. potential labourers. Awareness of land market determines the range over which landlords are able to perceive the market rents of neighbouring estates.

References